TSINGHUA UNIVERSITY
Department of Electronic Engineering
Probability and Stochastic Processes (1)
(Spring 2021)

Problem Set 14
Due: Jun. 7, 2021

1. Section 6.1 Problem 1.

   Each of n packages is loaded independently onto either a red truck (with probability p) or onto a green truck (with probability $1-p$). Let R be the total number of items selected for the red truck and let G be the total number of items selected for the green truck.

   (a). Determine the PMF, expected value, and variance of the random variable R.

   (b). Evaluate the probability that the first item to be loaded ends up being the only one on its truck.

   (c). Evaluate the probability that at least one truck ends up with a total of exactly one package.

   (d). Evaluate the expected value and the variance of the difference $R-G$.

   (e). Assume that $n \geq 2$. Given that both of the first two packages to be loaded go onto the red truck, find the conditional expectation, variance, and PMF of the random variable R.

2. Section 6.2 Problem 13.

   Transmitters A and B independently send messages to a single receiver in a Poisson manner, with rates of λ_A and λ_B, respectively. All messages are so brief that we may assume that they occupy single points in time. The number of words in a message, regardless of the source that is transmitting it, is a random variable with PMF

$$p_W(w) = \begin{cases} 2/6 & \text{if } w = 1 \\ 3/6 & \text{if } w = 2 \\ 1/6 & \text{if } w = 3 \\ 0 & \text{otherwise} \end{cases}$$

   and is independent of everything else.

   (a). What is the probability that during an interval of duration t, a total of exactly nine messages will be received?

   (b). Let N be the total number of words received during an interval of duration t. Determine the expected value of N.

   (c). Determine the PDF of the time from $t = 0$ until the receiver has received exactly eight three-word messages from transmitter A.

   (d). What is the probability that exactly eight of the next twelve messages received will be from transmitter A?

3. Suppose a bank has 2 tellers. When you enter the bank, you find that both tellers are busy serving other customers, and there are no other customers in queue. The service times of the two tellers are independent distributed exponential random variables, with parameters λ_1 and λ_2 respectively. What is the probability that you will be the last to leave?

4. ***Bonus.** Customers arrive at a certain retail establishment according to a Poisson process with rate λ per hour. Suppose that two customers arrive during the first hour. Find the probability that both arrived in the first 20 minutes.